

Public Economics (ECON 131)

Section #10: Social Insurance

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1 Key Concepts

1.1 Insurance Intro

- **Social insurance programs** are government interventions in the provision of insurance against adverse events.
 - Eg: health insurance, disability insurance, unemployment insurance
 - Government provision can address market failures. May also be motivated by desire to redistribute, provide services that are considered a basic right, address individual failures, or reduce administrative costs
- These insurance systems allow for consumption smoothing across states of the world.
- **Insurance premiums:** Money that is paid to an insurer so that an individual will be insured against adverse events.

1.2 Expected Utility Model

- Individuals maximize expected utility (EU) over possible states of the world, where $q = \text{Pr}(\text{adverse event})$:
- An insurance premium is **actuarially fair** if it is set equal to the insurer's expected payout.
- Asymmetric information: individuals have private information about their risk factors.
 - **Adverse selection**: individuals with higher risk are more likely to purchase insurance. This makes insurance costlier, and may make the market unravel
 - **Moral hazard**: individuals may change their behavior in response to being insured

2 Examples

2.1 Adverse Selection

Do health insurance and car insurance markets both suffer from adverse selection? Are there reasons why the government should intervene in the health insurance market and not in the car market?

2.2 Moral Hazard

Unemployment insurance in the US typically pays the unemployed 50% of their previous wage for about 6 months. During the recession of 2009 it was extended temporarily to a year. Why is unemployment insurance far from 100% replacement? Why did government decided to extend UI during the recession?

3 Practice: Disability Insurance

Consider an economy where there are three types of people who want to purchase disability insurance. Each type has the same health-based risks, such that they each have a 2 percent chance of being incapacitated due to health risks, which are uncorrelated with their risk taking behavior. However, each type of person differs in their hobbies and work in the following manner:

- Low-risk types walk to work and have very low-risk hobbies. Their outside risk of being incapacitated is 3 percent.
- Medium-risk types drive to work and actively play soccer on the weekends. Therefore, their non-health risk of being incapacitated is 8 percent.
- High-risk types bike to work, work as firefighters, and skydive on weekends. Their outside risk of being incapacitated is 58 percent.

Assume that there are equal numbers of each type. Long-term care insurance provides income if they are incapacitated for the rest of their life (there are no additional costs). Suppose individuals have the following utility function over consumption (or income):

$$u(c) = \log(c)$$

Individuals earn \$500 if healthy, but only \$10 if incapacitated. They make the decision to purchase insurance before the event occurs.

- (a) Show that with competitive insurance markets (and actuarially fair premiums) that any given individual will choose to fully insure.

- (b) What is the actuarially fair price for the insurance for each group? What is the expected utility of each group given that price if they buy insurance? What is the expected utility if they do not buy insurance? What prices are each groups willing to pay for the insurance? Assume insurers are perfectly competitive.

- (c) Show graphically how someone with a log utility (exhibiting diminishing marginal returns to consumption) can achieve a higher utility by insuring.

- (d) Now assume that the insurance company is not able to differentiate between the three types. Therefore, it offers a policy that fully insures individuals at the same price to all three types. Assuming all three types buy this insurance policy, what is the price for insurance?
- (e) Prove that with the price you found in part (d), the low-risk type would not be willing to buy the full insurance policy.
- (f) Since the low-risk type does not buy the policy, the insurance company cannot offer the policy at the price from part (c). What is the new price for this policy if the risk averse drop out of the market? Who will buy at this price?

(g) You have demonstrated an example of a market unraveling. Explain the intuition for why it happened.

(h) How would results change if the individuals' income is zero when incapacitated?